



Pre-Paid Automatic Control Filling System for Water Filling Stations

Powered by S!aP — connecting the field, CCTV, payment & network systems into one intelligence layer

One centralized platform for tanker filling, prepaid revenue, surveillance & network intelligence

Prepared for

Ministry of Electricity & Water (MEW)

Water Projects Sector · Al Dhaher Lorry Filling Station — State of Kuwait

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Patented Technology · ISO/IEC 27001 · IEC 62443-aligned · Glass Box Explainable AI

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1 Executive Summary

1.1 Project Context: One Centralized Platform for Al Dhaher & the Bulk Network

The Ministry of Electricity & Water (MEW) is modernising its Al Dhaher lorry filling station into an online, prepaid, account-based water-tanker charging system, fully integrated into a Remote Centralized Water Tanker Filling Management System. The station comprises 42 filling points/bays, each with custody-transfer metering, a tight shut-off control valve, an RTU and driver interface. Today filling is card-credit based; MEW needs a secure, centralized platform that unifies the 42 bays, the Local Control Centre, the central data centre at Salmiya, the Disaster Recovery centre at MEW Head Office (South Surra), IP video surveillance, network flow monitoring at Shuwaikh (WNCC) and the online payment gateway – one place for operations, revenue, surveillance and intelligence.

What MEW needs is not another isolated SCADA or billing system, but **an enterprise platform above the field, surveillance, payment and network systems** that aggregates all 42 bays into a single historian and command dashboard, runs the prepaid charging and K-net settlement, manages incidents and reporting, and applies AI analytics – served to operators at Al Dhaher and Salmiya, to tanker owners through the MEW Pay app, and to management as an executive cockpit, with a Disaster Recovery site for continuity.

1.2 The Proposed Solution: The "Glass Box" Centralized Platform

Astrikos proposes exactly that platform, built on **S!aP – Astrikos AI's patented operational intelligence platform**. Critically, **Astrikos does not supply, replace or modify the field instrumentation, valves, cameras or gateways**. The specialist field devices remain in place; S!aP connects to each bay RTU, camera and gateway, historizes every transaction and signal centrally, and delivers the centralized historian, command dashboard, prepaid billing, surveillance integration, reporting and AI analytics as the single group-wide platform. Because it is built on a Glass Box architecture, every dashboard, KPI, rule, model and report is transparent, auditable and configurable by MEW – MEW owns the data, the logic and the roadmap, with no black-box lock-in.

1.3 Strategic Pillars of the Technical Offer

Strategic Pillar	What it Means for MEW
Above the field systems, not replacing them	S!aP integrates the specialist hardware MEW requires (meters, valves, RTUs, cameras, gates)— no rip-and-replace, no field disruption, protecting MEW's investment.
One centralized truth	All 42 bays, the LCC, Salmiya, DR and WNCC aggregated into a single historian and command dashboard.

Secured revenue	Prepaid, account-based charging via K-net with immutable audit trail and automatic reconciliation to the MEW bank account.
Minimised water loss	Custody-transfer metering, mass-balance analytics and AI leak detection reduce non-revenue water.
Resilient & offline-capable	Dual-link mesh WAN failover; bays operate on last-known balance and store-and-forward when connectivity drops; DR site with continuous replication.
Open & sovereign, secure by design	IEC 62443 zones & conduits, ISO/IEC 27001 controls, multi-layer encryption – with full explainability and MEW ownership.

1.4 At a Glance

Requirement	Astrikos S!aP Response
Real-time acquisition from 42 bays + meters, valves, cameras, gates	S!aP Connect via bay RTUs (Modbus/OPC-UA), CCTV (RTSP), gateways (REST), Edge at Al Dhaher LCC
Central historian – 10-year+ retention	S!aP Datalake, open time-series DB, DR-replicated to South Surra
Prepaid charging, billing, K-net settlement, reporting	S!aP BPM + Datalake + Agentic – wallet, invoicing, reconciliation, shift/day/month reports
Centralized dashboard, surveillance, unlimited users	S!aP Viz & Astriverse – command dashboard, CCTV/LPR wall, MEW Pay & web portal
DR & business continuity	Geographically separate DR at MEW HO (South Surra), continuous replication (RPO ≈ 0)

2 About Astrikos & the S!aP Platform Philosophy

2.1 Corporate Overview: Deep Tech for Critical Infrastructure

Astrikos AI Technologies Limited is a deep-technology company specialising in operational intelligence for complex infrastructure – including utilities, water, smart cities, industrial & energy, data centres, airports and ports. Our patented S!aP platform bridges OT and IT, layering above existing control, metering, surveillance and payment systems to turn real-time operational data into centralized monitoring, revenue assurance, KPIs and AI analytics. With offices across India, the USA, the UAE (our MEA Regional Headquarters) and Singapore, and

delivery experience across the Middle East, Africa and Asia, Astrikos is well placed to serve MEW's operations in Kuwait.

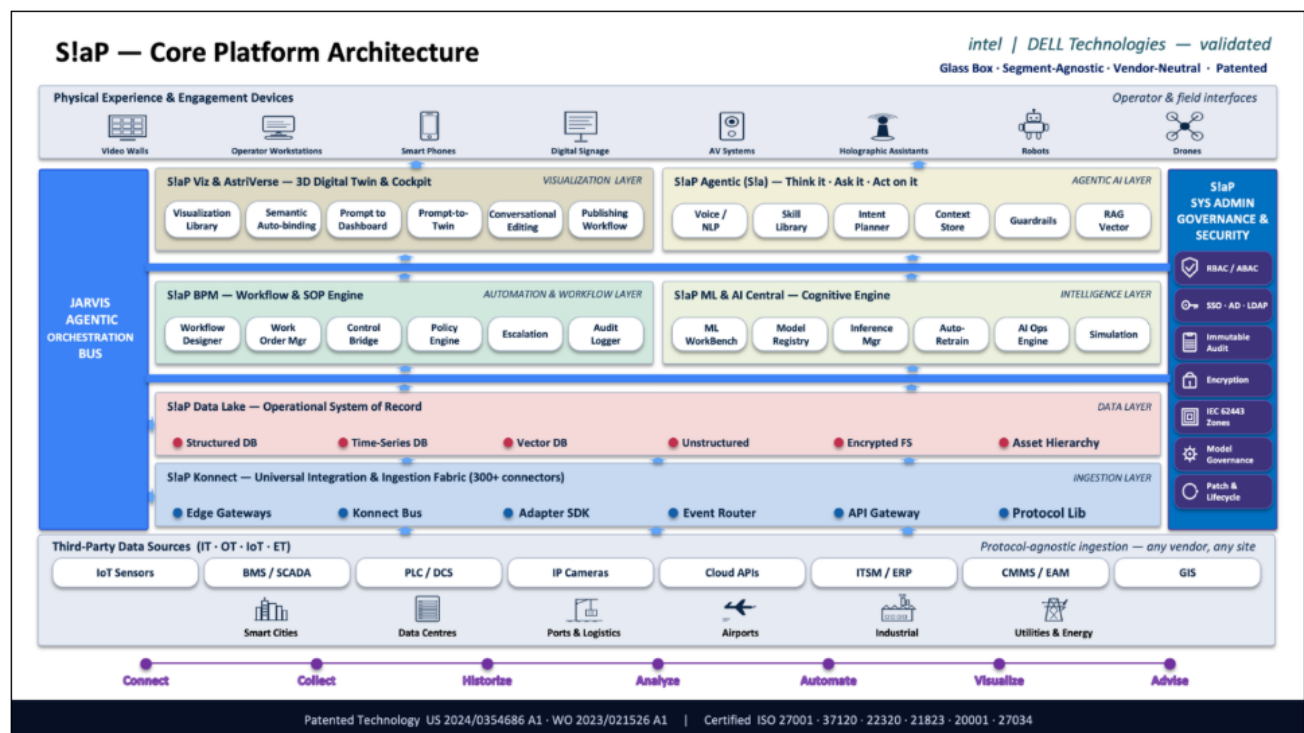
Astrikos holds patented technology status and aligns to ISO/IEC 27001:2013 (Information Security), ISO 22320:2018 (Emergency Management), ISO 37120/37122 (Smart City Indicators), ISO/IEC 21823-1:2019 (IoT Interoperability), ISO 20001-1:2018 and ISO/IEC 27034-1:2011, and follows IEC 62443 principles for OT security.

2.2 The S!aP Philosophy: A Layer Above, never a Replacement

S!aP never displaces the systems beneath it. The bay RTUs, meters and valves continue their local control and custody-transfer measurement unchanged; the cameras, LPR and gate barriers continue their function; S!aP reads above them and adds the centralized historian, prepaid revenue engine, analytics and dashboards. This is the essence of the platform: a vendor-neutral intelligence layer that ingests data from any OT/IT/IoT source without disrupting installed infrastructure, following the lifecycle Connect, Collect, Historize, Analyze, Integrate, Visualize, Advise.

3 The S!aP Solution Architecture: Six Pillars of Intelligence

S!aP delivers its capability through six integrated modules plus a cross-cutting governance layer. Each functional requirement in Section 5.1E maps directly to a S!aP pillar.



#	Module	Role in the MEW AI Dhafer Platform
1	S!aP Connect	Connects to the 42 bay RTUs (Modbus/OPC-UA), IP cameras (RTSP), gate barriers, network flow instruments and the K-net gateway; Edge ingestion at the AI Dhafer LCC over the secured WAN.
2	S!aP Datalake	Central historian – open time-series store, 10-year+ retention of every transaction and signal, DR-replicated to South Surra.
3	S!aP ML & AI Central	Water-loss / mass-balance analytics, predictive maintenance on valves & meters, LPR & leak detection, fraud and demand analytics – Glass Box explainable.
4	S!aP BPM	Prepaid charging logic, incident lifecycle, PIN-lockout & escalation SOPs, K-net settlement reconciliation, report scheduling and immutable audit.
5	S!aP Viz & Astriverse	Command dashboard, 3D digital twin, CCTV/LPR wall, executive cockpit; the MEW web payment portal and MEW Pay mobile app; unlimited web/intranet/LAN users.
6	S!aP Agentic (S!a)	Natural-language querying and automated shift/day/month reporting and advisory.
+	S!aP Core (Sys Admin)	RBAC, SSO/AD, immutable audit; supports the required operator and unlimited customer access.

Module 1: S!aP Connect (Field, CCTV, Gateway Connectivity)

S!aP Connect is the vendor-neutral integration layer. At AI Dhafer it connects to each of the 42 bay RTUs over Modbus/OPC-UA, to the IP cameras over RTSP, to the gate barriers and network-flow instruments, and to the K-net/MEW payment gateway over HTTPS. A local Edge at the Local Control Centre buffers data and transmits it to the Salmiya data centre over the secured WAN. Connect is northbound and read-only towards the field – it consumes values and does not write to, or alter, the RTUs, valves or control loops. Store-and-forward buffering ensures no data loss during any WAN interruption.

Module 2: S!aP Datalake (The Central Historian)

The Datalake is the central historian the tender requires. It stores every ingested transaction and signal from all 42 bays plus surveillance, gate and network-flow events in an open, high-performance time-series database with 10-year+ retention (configurable and extensible), organised by site, bay and device. Built on open technology, it avoids proprietary lock-in and is continuously replicated to the DR site at South Surra for business continuity.

Module 3: S!aP ML & AI Central (Water-Loss, Assets & Revenue Analytics)

ML & AI Central provides both the standard analytics and the AI differentiator. It runs custody-transfer mass-balance and water-loss analytics (inlet vs. dispensed), predictive maintenance on valves and meters, license-plate & facial recognition and leak detection on video, and fraud/anomaly and demand analytics. Under Glass Box governance, every KPI formula, threshold and model is visible, configurable and explainable by MEW's engineers.

Module 4: S!aP BPM (Prepaid Charging, SOPs & Reporting)

S!aP BPM runs the prepaid charging logic – validate account, dispense set volume, debit exact volume, issue SMS/email receipt – plus the incident lifecycle, PIN-lockout and escalation SOPs, and K-net settlement reconciliation to the MEW bank account. It orchestrates the shift/day/month production, revenue and performance reports on schedule and on demand, with an immutable audit trail of every transaction and action.

Module 5: S!aP Viz & Astriverse (Centralized Dashboards & Apps)

S!aP Viz delivers the centralized command dashboard hosted at Salmiya, serving operators at Al Dhaher and the Salmiya control centre, plus the CCTV/LPR wall, the executive cockpit, and role-based dashboards. It also delivers the MEW web payment portal ("Water Tanker Customer Services") and the MEW Pay mobile app (Android/iOS, English/Arabic). The Astriverse 3D digital twin provides spatial situational awareness of the station.

Module 6: S!aP Agentic (S!a) & S!aP Core

S!a ("Sia") is the conversational agentic AI – "Think it. Ask it. Act on it." Operators ask questions in plain language across operations, revenue, assets and security and receive explained, confidence-scored answers, and S!a automates the recurring shift/day/month reports with human-approval gates on any action. S!aP Core (Sys Admin) underpins the platform with RBAC, SSO/AD integration and immutable audit logging.

4 Project Understanding

4.1 Overview of the Requirement

MEW requires a prepaid automatic control filling system across 42 bays at Al Dhaher, integrated into a centralized management system at Salmiya with Disaster Recovery at South Surra, covering billing, invoicing, accounting, online payment, video surveillance, network flow monitoring and cyber security. Astrikos' scope is that centralized platform and the edge connectivity to the field, surveillance, gate and payment systems – not the specialist field hardware itself.

4.2 Scope Summary – Al Dhaher & Central

Requirement	Al Dhaher (Local)	Central (Salmiya + DR)
Filling bays	42 bays · custody-transfer EMF · tight-shut valve · RTU	Aggregated historian & dashboard
Authentication	QR / Barcode / RFID + PIN at each bay	Account & wallet management
Surveillance	UHD IP CCTV · LPR · leak detection · 60-day NVR	Integrated to Salmiya monitoring centre
Network flow	DN800 flow & pressure (solar)	Integrated to WNCC SCADA, Shuwaikh
Payment	—	K-net · wallet · invoicing · MEW bank settlement
Continuity	Offline store-and-forward (bay up to 15 days)	DR at South Surra · continuous replication

4.3 Functional Requirements – How S!aP Meets Them

- **Sequence of operation (scan/PIN → validate → autofill → debit → receipt):** S!aP BPM orchestrates the flow; Connect drives the RTU/valve/meter loop.
- **Prepaid, account-based charging & K-net settlement:** S!aP BPM + Datalake maintain the wallet; exact-volume debit; reconciliation to MEW bank.
- **Offline operation:** bay and concentrator run on last-known balance; store-and-forward on WAN loss (bay up to 15 days).
- **Surveillance, LPR & leak detection:** S!aP Connect (RTSP) + ML & AI; integrated to the Salmiya monitoring centre.
- **Historic reporting & analytics:** S!aP Datalake + ML & AI — shift/day/month reports, water-loss and demand analytics.
- **Remote lock-out & fraud control:** PIN-abuse lockout, plate/QR mismatch detection, remote enable/disable of bays.

4.4 Response to Tender Information Requests

The tender asks bidders for specific information. This offer answers each, as summarised below, with detail in the referenced sections.

Information Requested	Section	Summary
Solution architecture (meets §2–§4)	5	S!aP above the field; central Salmiya + DR; Edge over WAN
Infrastructure / hardware per tier	5	Indicative specs provided; confirmed at design
Data storage sizing (10-yr) + growth	5	Indicative estimate & method; scalable historian
Cyber security (IEC 62443 / ISO 27001)	5	Zones & conduits, encryption, SIEM/EDR, DR
Implementation & go-live timeline	7	Phased plan incl. 1-bay trial & 30-day reliability run
Licensing model & scalability	8	Modular, T-shirt sized; State-wide extensible
Commercial quotation (CAPEX / OPEX / AMC)	8	Structured; indicative placeholders [TBD]
Maintenance, support & SLA	8	Tiered AMC / SLA options; 2-year maintenance
Reference implementations	10	Utilities & industrial deployments (under NDA)

5 Solution Overview

5.1 Solution Architecture

The architecture follows the tender's network model. Each bay RTU, camera, gate and network-flow instrument at Al Dhaher is read by a local S!aP Connect Edge at the Local Control Centre, which transmits data to the Salmiya main data centre over the secured WAN. The Salmiya data centre hosts the full S!aP platform – the central historian, prepaid billing and analytics engine, alarm/incident and reporting engine, surveillance integration and the command dashboard – serving operators and unlimited customer users, with continuous replication to the DR site at MEW Head Office, South Surra.

MEW — Centralized Water Tanker Filling, Revenue & Intelligence on S!aP

S!aP connects to the field, CCTV, payment and network systems and delivers one centralized command, prepaid billing, surveillance & AI-analytics platform

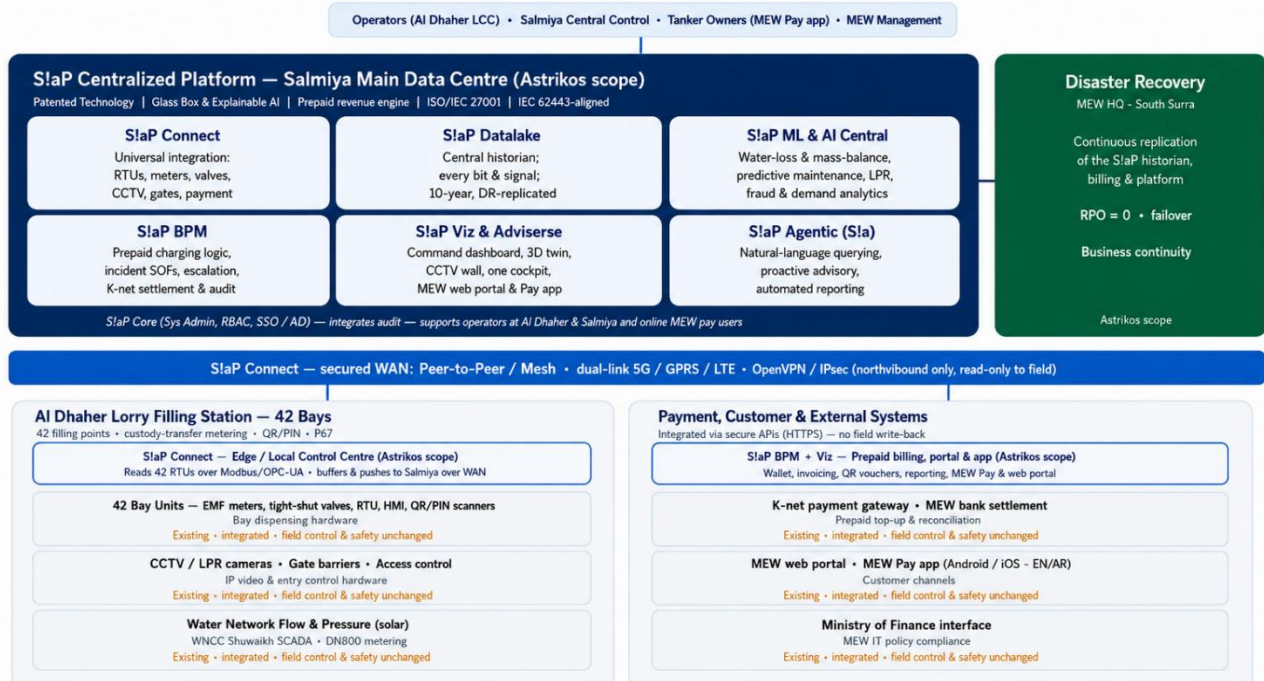


Figure 1 — S!aP as the centralized platform above the existing field, CCTV and payment systems; bay RTUs, meters, valves, cameras and gateways remain in place and are integrated, not replaced.

5.2 Scope Boundary — What Astrikos Supplies

In Astrikos scope: the S!aP platform at the Salmiya data centre (historian, prepaid billing/analytics, alarms, reporting, dashboards, surveillance integration, AI), the DR replica at South Surra, the S!aP Connect Edge at the AI Dhaheer LCC, the integration and tag/stream mapping to the 42 bay RTUs, cameras, gates, network-flow instruments and the K-net gateway, and the MEW web portal and MEW Pay app.

Integrated but not supplied by Astrikos: the specialist field hardware — electromagnetic flow meters, tight shut-off valves, RTUs, HMIs, scanners, IP67 enclosures, IP/PTZ/LPR cameras, NVRs, gate barriers, solar-powered network-flow instruments, servers and network switches — together with the K-net payment gateway. These are provided by the qualified system-manufacturer sub-contractor (per §2.9) and OEM partners and are integrated through S!aP Connect. Astrikos remains the single point of accountability for the integrated platform.

5.3 Connectivity & Data Flow

S!aP Connect reads the field over standard industrial protocols (Modbus/OPC-UA), video over RTSP and payment over HTTPS — so no change to the RTUs or control loops is required beyond enabling access for the Edge. The Edge applies change-based collection to minimise bandwidth, buffers locally (store-and-forward) to survive WAN outages without data loss, and

forwards to the central historian over a peer-to-peer / mesh WAN with dual-link (wired + 5G/GPRS/LTE) failover. All flow is northbound from the field to S!aP; the platform issues no control commands to the dispensing loop.

5.4 Infrastructure & Hosting

S!aP is a lightweight, containerised platform deployed on standard servers at the Salmiya data centre, with a mirrored DR environment at South Surra and continuous replication between them. Indicative infrastructure, sized for confirmation during detailed design, is:

Tier	Indicative Specification	Notes
Edge (Al Dhaher LCC)	Industrial server(s), 8–16 cores, 32–64 GB RAM, 2 TB SSD, redundant PSU	Field/CCTV/gateway ingestion + store-and-forward
Salmiya DC (main)	HA pair, 16–32 cores, 128 GB RAM, 16–32 TB usable (RAID), enterprise OS	Historian, billing, analytics, dashboard, surveillance; redundant
DR (South Surra)	Mirror of DC spec (single or HA), matched storage	Continuous replication target; failover

Final hardware and redundancy are confirmed at design against MEW standards; the platform is hardware-agnostic and can run on MEW-provided or Astrikos-specified infrastructure.

5.5 Data Storage Sizing (10-Year Historian) & Growth

Storage depends on the number of tags/streams per bay, the effective logging rate after change-based compression, transaction volume, and video retention. With 42 bays plus surveillance metadata, network-flow and payment records, a practical planning estimate is on the order of a few TB per year of compressed time-series and transactional data (excluding raw video, which is retained 60 days on the NVR tier). Astrikos recommends provisioning 16–32 TB usable at Salmiya (and matched capacity at DR) to cover 10-year retention plus headroom and indexing. The open historian scales horizontally, and retention, roll-up and archival policies are configurable — so storage grows predictably with bays, transactions and retention. Exact figures are confirmed once sampling rates and retention per stream are agreed during design.

5.6 Security Architecture

S!aP follows IEC 62443 zone-and-conduit segmentation (-4-1/-4-2/-4-3) and ISO/IEC 27001. Components are distributed across segmented zones: an Enterprise (Web) zone for the portal, app and APIs; a DMZ/Extractor zone that sanitises incoming data; and an Operations zone housing the Core and Datalake, isolated from internet-facing networks. The Edge sits at the

station boundary as a read-only client, so the dispensing control and safety layers are never exposed or altered through the platform. AES-256 / TLS 1.3 encryption, OpenVPN/IPsec tunnels, certificate-based authentication, SIEM, EDR, patch management, whitelisting, RBAC, SSO/AD, immutable audit and DR replication complete the posture, supported by a documented cyber risk & vulnerability assessment.

6 Dashboards, KPIs & User Experience

This section shows what MEW's users see. Every view is centralized at Salmiya, spans the whole station, is role-based, and is served to operators at Al Dhaher and Salmiya and to tanker owners through the MEW Pay app. The screens below are from a working demonstration of S!aP configured for Al Dhaher; views are fully configurable to MEW's assets and KPIs.

6.1 Secure Access & Unlimited Users

Access is governed by S!aP Core: role-based access control for operators, engineers and management, SSO/Active Directory integration, and an immutable audit trail. The web dashboard and MEW Pay app serve the required operator and unlimited customer users, with no per-user viewing constraint.

6.2 Central Command Dashboard

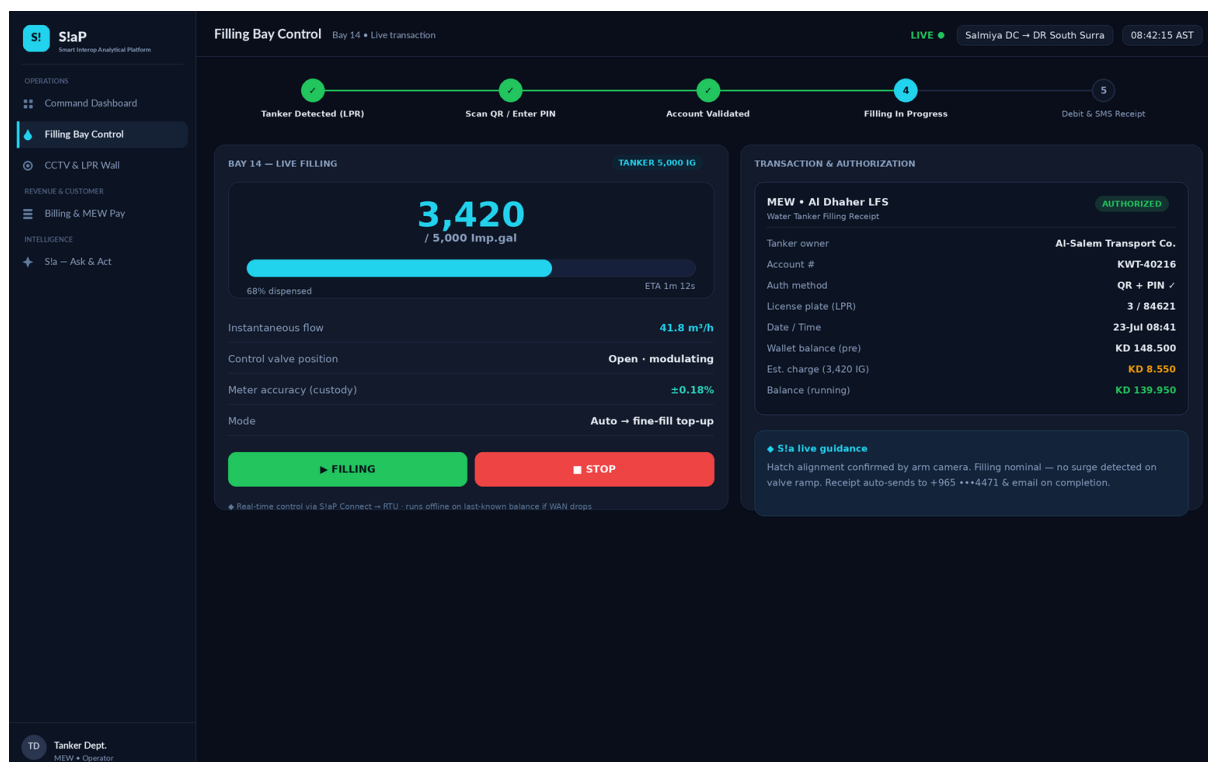
A single live picture of all 42 bays, station inlet flow, water dispensed, revenue, tankers served and open alarms, with predictive advisories and cyber-security posture in real time. Drill-down moves from station, to bay, to device and transaction. (RFP \$2.2, \$2.3, \$3.2, \$4.0)



SlaP Command Dashboard – 42-bay status, inlet flow gauge, hourly volume/revenue with Sla demand forecast, alarms and IEC 62443 / ISO 27001 posture.

6.3 Filling Bay Control & Transaction Flow

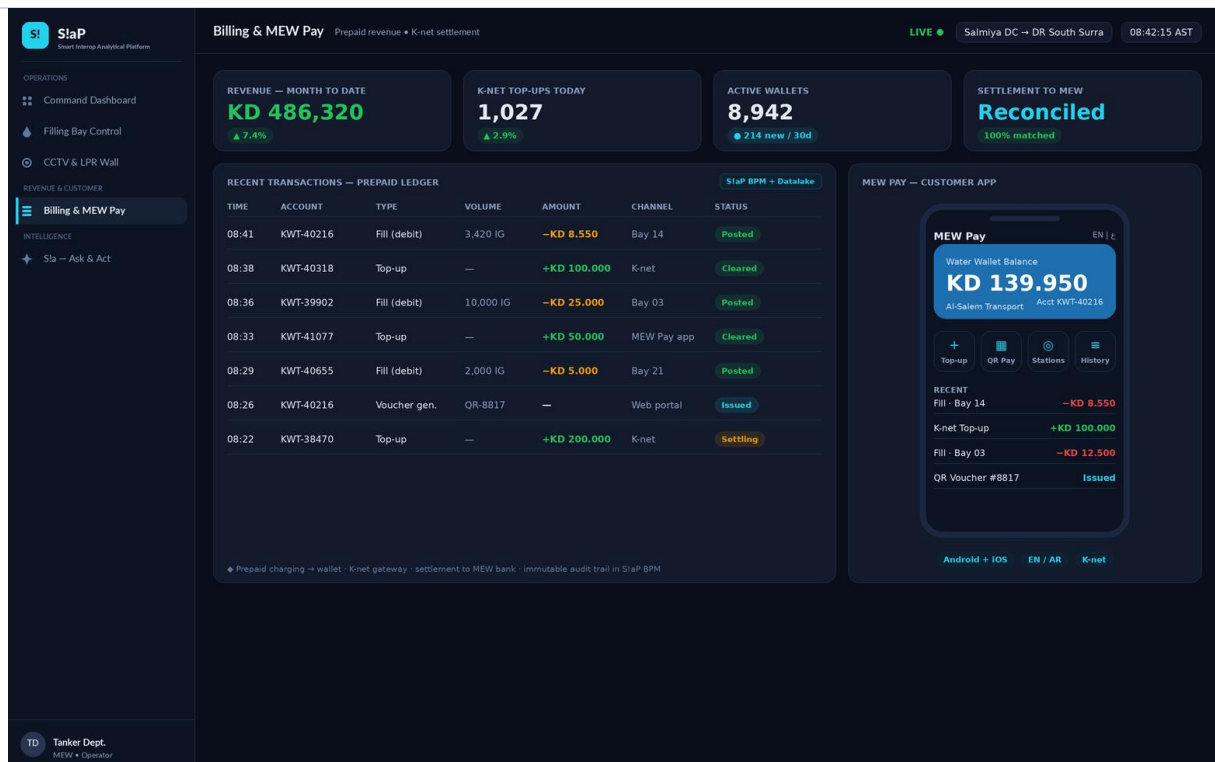
Each bay follows the tender's sequence of operation: LPR detection, QR/PIN authentication, account validation, automatic volume-controlled filling with fine-fill top-up, exact-volume debit and SMS/email receipt – continuing on last-known balance if the WAN drops. (RFP \$2.1, \$3.5, \$4.2)



SlaP Filling Bay Control — live fill against set-point, custody-grade metering, valve modulation, prepaid authorization and Sla hatch-alignment guidance.

6.4 Billing, Prepaid Wallet & MEW Pay

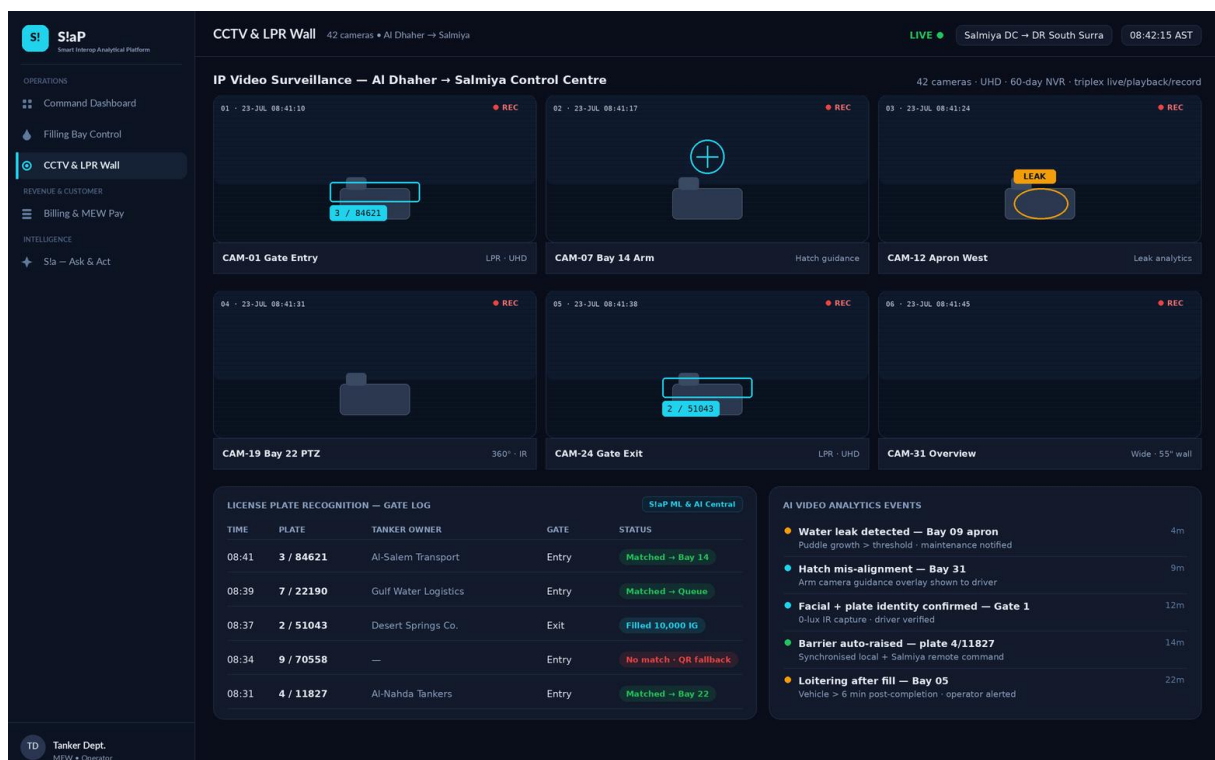
The prepaid ledger records every top-up and fill against the tanker owner's wallet, with K-net top-ups, QR vouchers and automatic reconciliation to the MEW bank account. The MEW Pay app (Android/iOS, EN/AR) provides registration, wallet management, quick payment, station locator and statements. (RFP \$2.3, \$2.4, \$2.5)



SlaP Billing & MEW Pay — prepaid transaction ledger, K-net settlement status and the customer mobile application with wallet and QR payment.

6.5 CCTV & License Plate Recognition

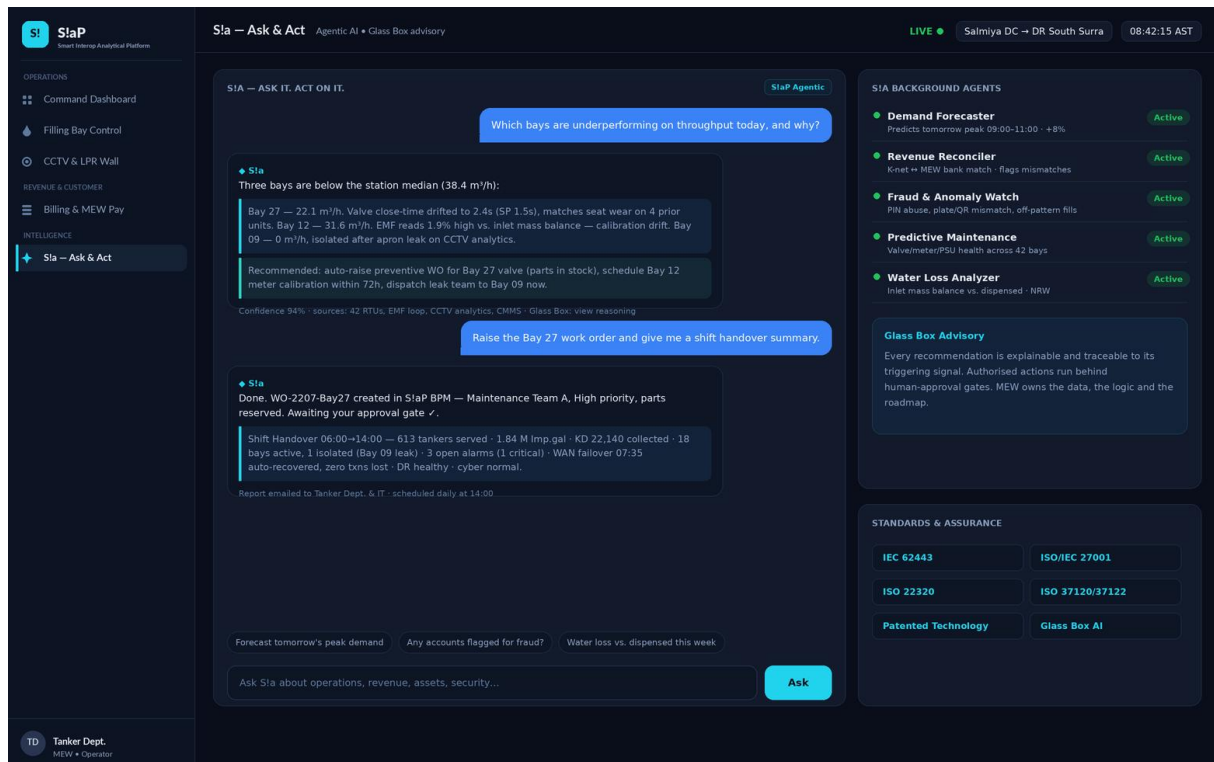
UHD IP surveillance with triplex live/playback/record and 60-day retention. AI video analytics deliver license-plate & facial recognition at the gates, arm-camera hatch guidance and water-leak detection — integrated to the Salmiya monitoring centre. (RFP \$2.6, \$2.7)



SlaP CCTV & LPR Wall – camera grid with plate/leak overlays, gate LPR log matched to accounts, and AI video-analytics events.

6.6 S!a – Ask it. Act on it.

S!a lets operators query operations, revenue, assets and security in plain language and receive confidence-scored, fully explainable advice. Background agents run 24/7 for demand forecasting, revenue reconciliation, fraud watch, predictive maintenance and water-loss analysis, with human-approval gates on any action. (RFP \$2.3, \$3.0)



S!a agentic interface – natural-language operations querying, Glass-Box recommendations, background agents and standards assurance.

6.7 Reporting – Shift, Day & Month

S!aP generates the shift-wise, day-wise and month-wise dispensing, revenue and performance reports the tender requires, on schedule and on demand, in MEW's required formats. Because reports are generated from the central historian, figures reconcile across bays, the station and time periods, and across the K-net settlement to the MEW bank account.

6.8 Configurability & Extensibility

Under Glass Box governance, MEW configures dashboards, KPIs, charging rules, alarm rules, reports, models and roles without vendor intervention. New bays, devices, or additional bulk filling stations across Kuwait are onboarded through configuration – so the platform extends State-wide over time without architectural redesign, protecting the investment.

6.9 Summary

The result is a single, centralized window into the whole operation – real-time monitoring, prepaid revenue, surveillance, reporting and AI analytics – delivered above the existing field, CCTV and payment systems, owned and configurable by MEW, and served with full DR resilience.

7 Implementation & Transition Strategy

7.1 Implementation Strategy

Delivery is structured to stand up the central platform quickly and onboard the station with minimal disruption – no changes to the RTUs or dispensing control beyond enabling access. In line with §2.9, one filling bay is commissioned and run for one month as a trial to verify functionality before roll-out, followed by a 30-day reliability trial run of the full station.

Phase	Time Duration	Key Activities	Deliverables
Phase 1 – Discovery & Design	Week 1-3	Confirm interfaces, tags, KPIs, charging rules, site requirements and system design.	Approved design documents, sizing, drawings and test plans.
Phase 2 – Platform & DR Build	Week 3-7	Deploy the central platform and DR site, configure HA, replication, security and integrations.	Operational primary and DR platforms with completed FAT.
Phase 3 – Pilot & Rollout	Week 8-14	Test one pilot bay, install Edge, connect 42 bays and validate dashboards, alarms and charging.	Pilot approval, full station integration and validated reports.
Phase 4 – Validation & Go-Live	Week 15-19	Complete testing, DR failover, training, go-live and maintenance handover.	SAT approval, trained users, go-live and support transition.

7.2 Indicative Timeline

An indicative timeline to full go-live is on the order of six months from contract, subject to confirmation at design (field access, WAN, hardware provisioning and MEW resource availability). A detailed, milestone-based schedule accompanies the final proposal, aligned to the Section 5.4 BOQ.

7.3 Scalability & Future Expansion

The platform scales by configuration – additional bays, devices, or further bulk filling stations across Kuwait – and its analytics improve over the asset life. MEW retains full ownership of data, historian, dashboards, models and roadmap under the Glass Box model, enabling a State-wide water-tanker intelligence programme on one foundation.

8 Scope of Work, Assumptions & Exclusions

8.1 Scope of Work

Astrikos' scope is the design, supply, configuration, integration, testing, commissioning, training and support of the centralized S!aP platform, comprising:

- S!aP platform deployment at the Salmiya data centre (central historian, prepaid billing/analytics, alarm/incident and reporting engine, surveillance integration, command dashboard) with high availability.
- A geographically separate DR environment at South Surra with continuous replication of the platform, historian and billing.
- A S!aP Connect Edge at the Al Dhaher LCC, integrated to the 42 bay RTUs, cameras, gates, network-flow instruments and the K-net gateway, with stream mapping and store-and-forward over the WAN.
- Configuration of the command dashboard, prepaid charging, KPIs, alarm rules, and shift/day/month reports; the MEW web portal and MEW Pay app.
- AI analytics – water-loss, predictive maintenance, LPR/leak, fraud and demand – under Glass Box governance.
- RBAC/SSO, unlimited-user access, security configuration (ISO/IEC 27001, IEC 62443-aligned), training, commissioning, the 1-bay trial and 30-day reliability run, go-live and 2-year maintenance.

8.2 Assumptions

- The bay RTUs expose accessible interfaces (Modbus/OPC-UA) with the required parameters, and access is enabled for the S!aP Edge.
- The specialist field hardware (meters, valves, RTUs, cameras, gates, solar network-flow, servers, switches) is supplied and installed by the qualified system-manufacturer sub-contractor and OEM partners per §2.9.
- MEW provides the Salmiya DC and South Surra DR facilities and either provides infrastructure to the agreed sizing or procures the specified hardware.
- K-net gateway access and MEW bank/Ministry of Finance interfaces are made available; tag/stream lists and charging rules are confirmed during design.
- MEW engineering and IT resources are available for design, access enablement, validation and acceptance.

8.3 Exclusions

- Supply, modification or maintenance of the field instrumentation, valves, RTUs, cameras, NVRs, gate barriers, solar network-flow hardware or servers, except where explicitly included in the commercial schedule.
- Any control, command or write-back to the dispensing loop – the platform is read-only and analytics/billing-only towards the field.
- Civil works, ducts, trays and cabling (per §2.8) unless explicitly included.
- Third-party licences (e.g. OS, database where MEW-standardised) and the K-net/bank transaction fees unless explicitly included.

9 Reference Implementations

Astrikos and S!aP are deployed across utility, industrial and infrastructure environments where the platform layers above existing control, metering, surveillance and payment systems to deliver centralized monitoring, revenue assurance and AI analytics. Relevant references in water/utilities and revenue-assurance contexts – including OPC/Modbus integration to existing field systems, prepaid billing and multi-site aggregation – will be shared under NDA with the formal proposal, with contactable client references on request.

10 Why Astrikos

- **Above the field, not a replacement** connects via standard protocols; RTUs, valves, cameras and gateways untouched; no rip-and-replace.
- **One platform, whole operation:** a single central historian, dashboard, prepaid revenue engine and surveillance view across the station, DR-resilient.
- **AI, not just trends:** water-loss, predictive maintenance, LPR/leak and fraud analytics under explainable Glass Box governance.
- **Open & sovereign:** open historian, no per-tag lock-in, unlimited users; MEW owns data, logic and roadmap; State-wide extensible.
- **Certified & proven:** patented technology, ISO/IEC 27001 and IEC 62443-aligned, with regional delivery experience.

11 Commercial Structure, Licensing & Support

SlaP license and Platform Implementation Services For the Simplex option				
All Prices in USD	Qty	Unit Price in USD	Total Price in USD	Remarks
SlaP License - AAI-SlaP-LIC Perpetual SlaP License includes • SlaP Konnect • SlaP Datalake • SlaPBPM • SlaP ML & AI Central • SlaPViz • SlaP Agentic	1	112,000.00	112,000.00	
SlaP Installation and Configuration Integration services: • Platform Installation • Platform Configuration • Sub-System Integration • Users and Security Configuration • Dashboard building	1	90,000.00	90,000.00	
Total in USD			\$ 202,000.00	
SlaP license and Platform Implementation Services for the Duplex option				
All Prices in USD	Qty	Unit Price in USD	Total Price in USD	Remarks
SlaP License - AAI-SlaP-LIC Perpetual DC & DR SlaP License includes • SlaP Konnect • SlaP Datalake • SlaPBPM • SlaP ML & AI Central • SlaPViz • SlaP Agentic	2	140,000.00	140,000.00	
SlaP Installation and Configuration Integration services: • Platform Installation • Platform Configuration • Sub-System Integration • Users and Security Configuration • Dashboard building	1	90,000.00	90,000.00	
Total in USD			\$ 230,000.00	

SlaP support and Maintenance	1 st Year	2 nd year	3 rd year	Total
Yearly Support Charges	24,000.00	25,200.00	26,460.00	75,660.00
Total in USD				\$ 75,660.00

Hardware Price

Description	Qty.	Unit Price in USD	Total Price in USD
Processor, 2 × Intel Xeon Gold 5418Y (12 Cores, 2.0 GHz) or 2 × Intel Xeon Gold 6526Y (12 Cores, 2.8 GHz)"24 Physical Cores" 64 GB DDR5 ECC RDIMM (2× 32 GB, expandable) RAM, 1 × 2 TB Enterprise SSD (RAID-1) and 4 × 4 TB Enterprise HDD (RAID-1), A 100 GPU	2	21,800.00	43,600.00
Total			\$ 43,600.00

Offer Validity: This offer shall remain valid until 31 August 2026.

Payment Terms Milestone

Payments Milestone – I

License Delivery: 100% of the licensing cost should be paid after the delivery of the product licenses.

Milestone – II

Services

- Mobilization advances -20%
- Platform installation and commissioning – 20%
- Subsystem Integration and customization – 50%
- UAT and Go-Live – 10%

Note: For Hardware Supply should be 50% advance along with PO and 50% against delivering the hardware

Astrikos AI Technologies Limited

connect@astrikos.ai · www.astrikos.ai · Think it. Ask it. Act on it.

Commercial-in-Confidence. The SlaP platform embodies patented technology and proprietary methods of Astrikos AI.